Bioinformatics assignment 2

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Part A, answering questions:

I. Number of reads in the fastq file: 2421219

II. The identifier, quality, and sequence of the first read:

ID: SRR36584168.1

Sequence:

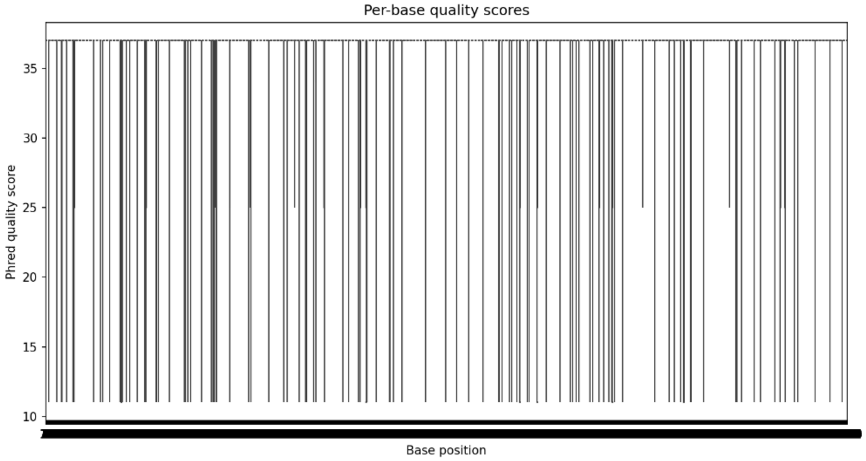
CGTTCATGACCGAAGCGGCGATCAACTCGCTGGCCGGCGAAGCGCTGAACGACATGTATGCGCAGTGCCAAACCAGCACTGGTCTGAATCCGGCATTCCAGGGGAATGGCGCTCAGAGTGAAATCCTTTCTATGGAGGCTCCTGAATAATGGCTCTCGCACCTCGTTTCCATACCGTAATCGCGGGCCCGGCCCGCAAGAATGACCCGCAGGTCATTGAAGCAATCATGGCGGCGGCAGTGAAGCCTGGGT

Quality: [37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 25, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 25, 37, 37, 11, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 25, 37, 37, 37, 37, 37, 37, 37, 37, 37, 25, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37]

III. The number of times the GCGATCAACTCGCTGG subsequence reapers: 7499

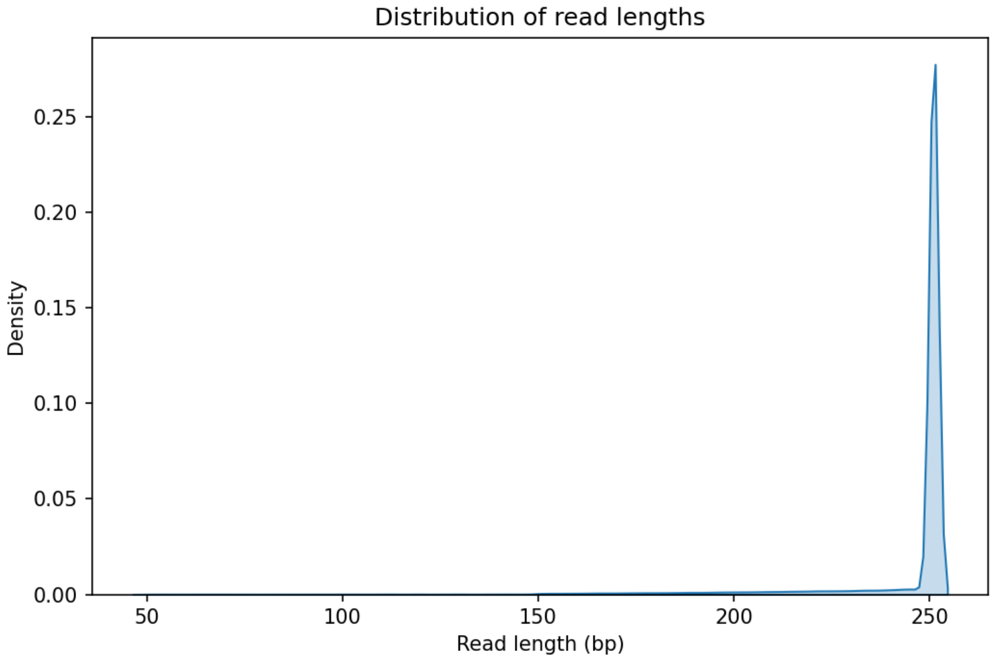
IV. Saved.

V. The plot:



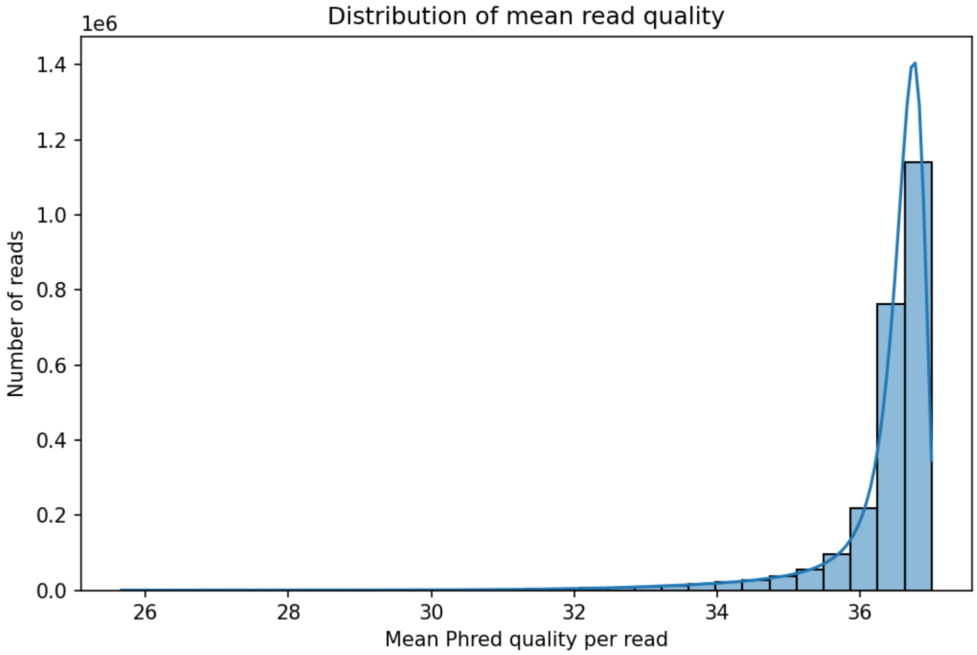
VI. Distribution read:

The

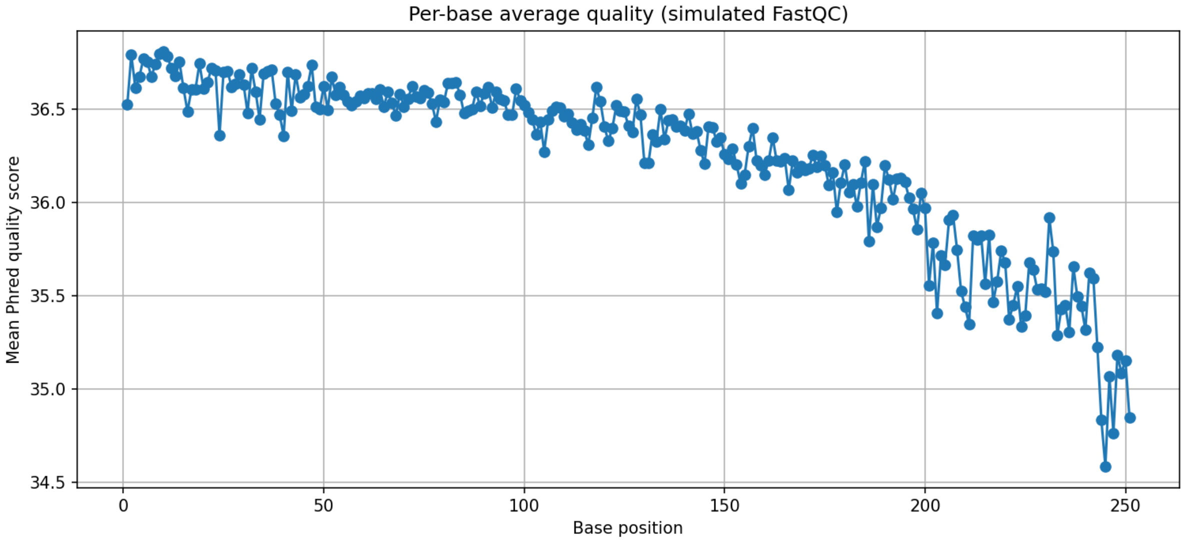


- **X-axis**: Read length (base pairs, bp)
- **Y-axis**: Density
- **Key observations**:
  - The distribution is highly concentrated around 250 bp, with nearly all reads falling within this length.
  - Very few reads are shorter than 200 bp.
- **Interpretation**:
  - This pattern is characteristic of paired-end Illumina sequencing with fixed read length.
  - The uniformity indicates excellent library preparation quality with minimal fragmentation variability or adapter contamination.
  - Consistent read lengths simplify downstream analyses such as alignment and assembly.

The density plot of read lengths shows a single dominant peak, indicating that the reads have a uniform length. This suggests consistent sequencing and no significant truncation of reads during preprocessing.



- **X-axis**: Mean Phred quality score per read
- **Y-axis**: Number of reads (scaled in millions)
- **Key observations**:
  - Strong Gaussian-like distribution peaking at Q36–Q37.
  - Very few reads fall below Q30, almost no low-quality outliers.
- **Interpretation**:
  - Phred scores ≥30 represent ≥99.9% base call accuracy; Q36 corresponds to 99.975% accuracy.
  - The high concentration of reads in the Q36–Q37 range indicates excellent overall data quality.
  - Suitable for sensitive applications such as variant calling, SNP detection, and de novo assembly without extensive filtering.



- **X-axis**: Nucleotide position within the read (1–250)
- **Y-axis**: Mean Phred quality score at each position
- **Key observations**:
    - Quality starts high (36.6) at the beginning of reads.
    - Remains relatively stable through position 150.
    - Shows a gradual decline from position 150 onward, reaching 34.5 at the 3' end (position 250).
- **Interpretation**:
    - This pattern is typical of Illumina sequencing chemistry, where signal decay, phasing/pre-phasing effects, and incomplete nucleotide incorporation accumulate toward read ends.
    - Although the drop is modest (only 2 Phred points), for high-precision applications, 3' end trimming may be advisable.
    - Overall, the per-base quality remains well above Q30 across the entire read length, indicating robust sequencing performance.

Since FastQC is not available, we performed quality control in Python. The per-read mean quality distribution shows that most reads have Phred scores above 30, indicating high-quality sequencing. The per-base quality plot shows a slight decrease at the 3′ ends, which is expected for Illumina sequencing. Overall, the dataset is suitable for downstream analysis.